

AWS Task-3

Task Description:

1. Create a S3 bucket, with no public access and upload files to the bucket & view the logs using cloudwatch for the uploaded files.
2. Launch two ec2-instances and connect it to a application load balancer, where the output traffic from the server must be an load balancer IP address

Techstacks needs to be used :

- AWS S3
- AWS EC2
- AWS LoadBalancer

How do I submit my work?

- Push all your work files to GitHub (O/P screenshot images must).
- Submit your URLs in the portal.
-

Terms and Conditions?

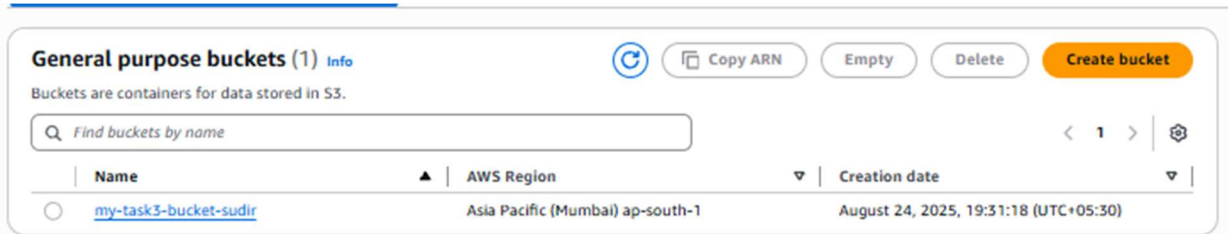
- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.
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NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

S3 Bucket with Server Access Logging

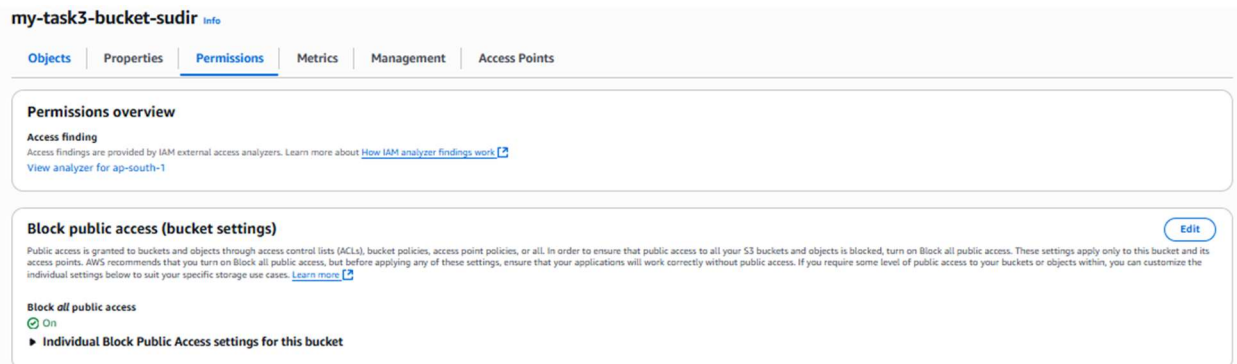
Go to **AWS Console** → **S3** → **Create Bucket**

- Enter bucket name → e.g., my-task3-source-bucket
- **Block Public Access** → keep **all 4 options checked** (default)
- Leave other options default → Create bucket



Again, go to **Create Bucket**

- Enter bucket name → e.g., my-task3-logs-bucket
- Block public access (keep it secure)
- Create bucket



Go to your **source bucket** (my-task3-source-bucket)

- Click **Properties** → **Server Access Logging** → **Edit**
- Select **Enable**
- Choose **destination bucket** → s3-task-access-logs

- (Optional) Add prefix like logs/ to keep logs organized
- Save changes

Server access logging

Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)

Server access logging
Enabled

Destination bucket
[s3://s3-task-access-logs](#)

Log object key format
[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

Edit

Open my-task3-source-bucket

- Click **Upload** → **Add Files**
- Upload a test file (e.g., index.html.txt)
- Confirm upload

Objects | Properties | Permissions | Metrics | Management | Access Points

Objects (2)

[Copy S3 URI](#)
[Copy URL](#)
[Download](#)
[Open](#)
[Delete](#)
[Actions](#)
[Create folder](#)
[Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	index.html.txt	txt	August 24, 2025, 19:38:12 (UTC+05:30)	0 B	Standard
☐	serverlogvalidation.txt	txt	August 24, 2025, 19:38:12 (UTC+05:30)	0 B	Standard

Logs may take **5–15 minutes** to appear.

- Go to my-task3-logs-bucket
- Look inside the folder (e.g., logs/)
- You will see log files like:

s3-task-access-logs Info

Objects | Properties | Permissions | Metrics | Management | Access Points

Objects (2)

[Copy S3 URI](#)
[Copy URL](#)
[Download](#)
[Open](#)
[Delete](#)
[Actions](#)
[Create folder](#)
[Upload](#)

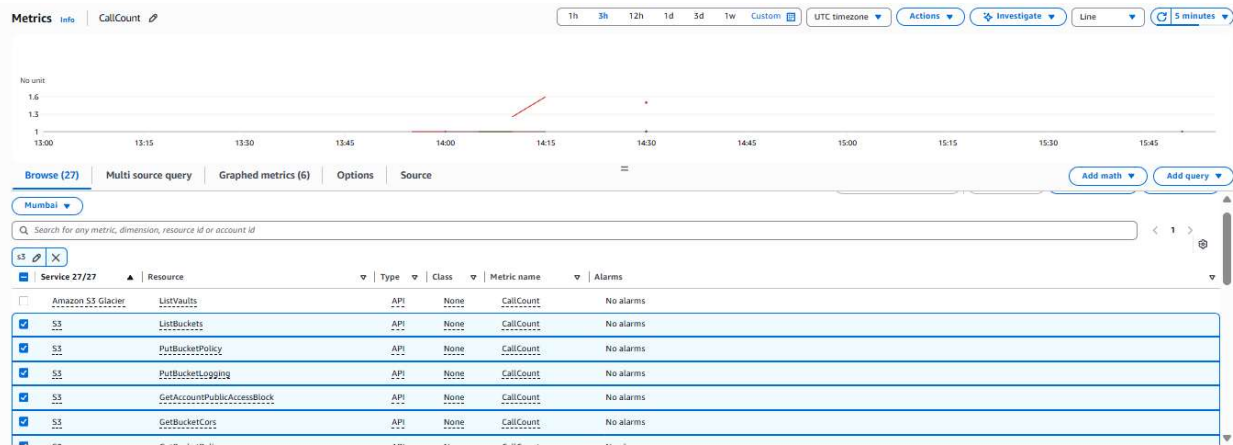
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Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	2025-08-29-14-20-52-DDAE044BA1B47180	-	August 29, 2025, 19:50:53 (UTC+05:30)	1.5 KB	Standard
☐	2025-08-29-14-21-13-E46510E62FC88006	-	August 29, 2025, 19:51:14 (UTC+05:30)	1.2 KB	Standard

Also we can try with cloud trail (Logging enabled) or

you can see api call count in cloud watch metric



EC2 + Application Load Balancer

Steps:

1. Launch **two EC2 instances** (Amazon Linux)

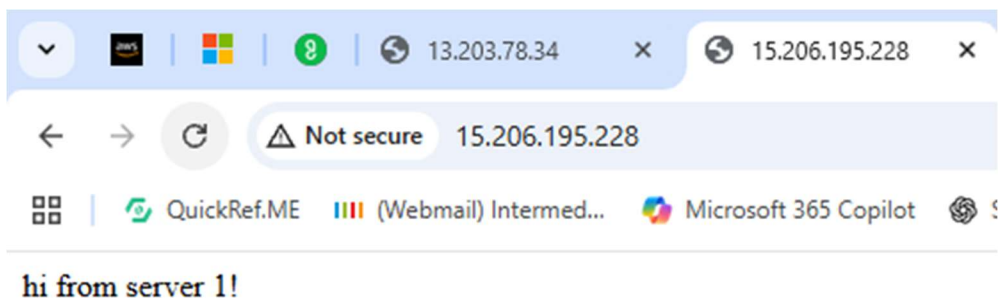
- Install a simple webserver (Apache/NGINX/) with unique index.html showing which server is serving the request.
- Example:
 - EC2-1 page → “Hello from Server-1”
 - EC2-2 page → “Hello from Server-2”

The screenshot shows the AWS Management Console 'Instances' page. It displays two EC2 instances, 'webserver-1' and 'webserver-2', both in a 'Running' state. The table includes columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, Public IPv4 DNS, and Public IPv4 address.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
webserver-1	i-0e9e3001fdff419	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-15-206-195-228.ap...	15.206.195.228
webserver-2	i-080ede783923f781e	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-13-203-78-34.ap-s...	13.203.78.34

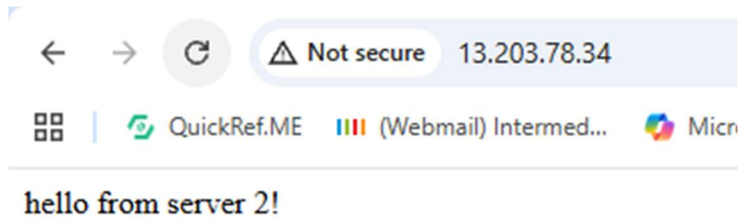
Server 1

```
[root@ip-172-31-1-2 ~]# history
 1  clear
 2  sudo yum install httpd -y
 3  sudo systemctl start httpd
 4  sudo systemctl enable httpd
 5  sudo systemctl restart httpd
 6  vim /var/www/html/index.html
 7  sudo systemctl restart httpd
 8  sudo systemctl status httpd
 9  history
[root@ip-172-31-1-2 ~]# cat /var/www/html/index.html
hi from server 1!
[root@ip-172-31-1-2 ~]#
```



Server 2

```
[root@ip-172-31-12-64 ~]# cat /var/www/html/index.html
hello from server 2!
[root@ip-172-31-12-64 ~]# history
 1  sudo yum install httpd -y
 2  sudo systemctl enable httpd
 3  sudo systemctl start httpd
 4  vim /var/www/html/index.html
 5  cat /var/www/html/index.html
 6  history
[root@ip-172-31-12-64 ~]#
```

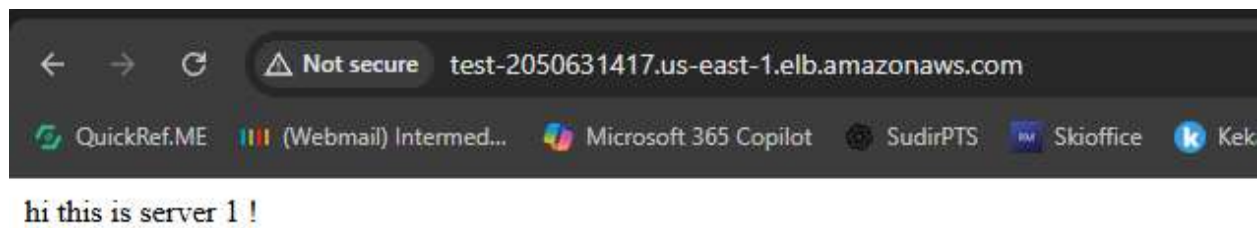


2. Create an **Application Load Balancer (ALB)**

- Add both instances in a **Target Group**
- Attach to ALB

3. Test from browser/curl → ALB DNS name will distribute traffic between servers

TRAFFIC ROUTED TO SERVER 1



TRAFFIC ROUTED TO SERVER 2

